METRIC

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MILITARY SPECIFICATION

BLASTING MACHINE, AUXILIARY FIRING, EMERGENCY

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is applicable for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a blasting machine, auxiliary firing which produces an electric output through generator action for emergency firing of fire power on full tracked combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2590

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SPECIFICATIONS

FEDERAL

TT-E-489	- Enamel, Alkyd, Glass, Low Voc Content.

MILITARY

MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-L-52043	- Lacquer, Semi-Gloss, Cellulose Nitrate.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.

DRAWINGS

ARMY

8675771	- Blasting Machine, Basic Unit.
11599607	- Blasting Machine, Basic Unit.
11637536	- Blasting Machine.
11679045	- Blasting Machine Assembly.
12282970	- Blasting Machine Assembly.

(Unless otherwise indicated, copies of the Federal and Military specifications, standards, and handbooks are available from the Navy Publication and Printing Service Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL BUREAU OF STANDARDS

Handbook H28	- Screw-Thread Standards for Federal Services.

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3).

3.2 Materials. Materials specified herein and in the referenced drawings, specifications, and standards shall be free from all defects and imperfections that might affect the serviceability and function of the finished product (see 4.6.1).

3.2.1 Recycled, virgin, and reclaimed material. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.6.1).

3.2.2 Parts. Functional parts exposed to the environment shall be fabricated from suitable corrosion resistant material or treated to prevent corrosion (see 4.6.1).

3.2.3 Threads. All internal and external threads shall conform to handbook H28 (see 4.6.1).

3.3 Design and construction. The design, construction and assembly of the blasting machine shall conform to drawing 8675771, 11599607, 11637536, 11679045, or 12282970 (see 4.6.1).

3.4 Performance.

3.4.1 Electrical output. The assembly shall have the electrical output listed in table I at each of the conditions specified (see 4.6.4.1 and 4.6.4.1.1).

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TABLE I. Electrical Output (not applicable to PIN 12282970).

CONDITION	ROTOR SHAFT INPUT 1/ TORQUE (N M) ± 10%	RESISTIVE LOADS (OHMS)2/	PEAK OUTPUT VOLTAGE (MINIMUM)
A	7.9	1.47 ± 1%	3.1 3/
B	7.9	20.50 ± 1%	27.0
C	7.9	Open Circuit	44.0
D	11.3	1.47 ± 1%	3.8
E	11.3	20.50 ± 1%	38.7
F	11.3	Open Circuit	60.0
G	14.7	1.47 ± 1%	4.4
H	14.7	20.50 ± 1%	49.4
I	14.7	Open Circuit	74.0

1/ Specified torque is the applied torque at the start of the stroke measured after 5° of shaft rotation when the blacklash of the gear drive mechanism is eliminated. Torque should be sufficiently constant throughout the stroke so that neither maximum nor minimum values of torque deviate from average torque by more than 15%. Torque at no point in the power stroke may exceed the maximum specified value of torque.
2/ Pure resistive load with no more than 1% inductive impedance component at 1000 Hz.
3/ Shall be 0.9 volts minimum 50 milliseconds after peak voltage.

3.4.1.1 Operating position. The assembly shall meet the electrical output requirements of 3.4.1 when operated in any of the following positions (see 4.6.4.2):

a. Shaft in a vertical position with handle down.
b. Shaft in a vertical position with handle up.
c. Shaft in a horizontal position.

3.4.1.2 Dielectric strength. The assembly shall show no evidence of loosening, cracking, charring, arc over, or other damage after being subjected to a test of 500 volts root mean square (rms) at a frequency of 60 hertz (Hz) applied for one minute between case and output leads (see 4.6.4.3).

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3.4.1.3 Insulation resistance. The assembly shall have an insulation resistance of not less than 100 megohms between the internal wiring and the case (see 4.6.4.4).

3.4.2 Physical characteristics.

3.4.2.1 Rotation. The direction of rotation of the handle to produce the electrical output (see 3.4.1) shall be clockwise when viewed from the handle end. One cycle of rotation shall be 150 ± 5.0 degrees and return to original position (see 4.6.5.1).

3.4.2.2 Torque. The assembly shall show no evidence of failure, electrical or mechanical, after repeated operation at 14.7 newton-meter (n m) of torque throughout the complete rotation of the rotor shaft (see 4.6.5.2).

3.4.2.3 Operational stability. The assembly shall withstand 10 000 cycles of operation at a rate not in excess of 10 cycles of rotation per minute. The torque, load resistance, and electrical output shall conform to condition B listed in table I (see 4.6.5.2).

3.4.3 Environmental requirements.

3.4.3.1 Waterproofness. The assembly shall be capable of withstanding 41 kPa differentialwhile submerged in water for 30 minutes, in the non-operating mode, with no evidence of leakage. The assembly shall be electrically and mechanically operable subsequent to submission. The electrical output shall conform to 3.4.1 (see 4.6.6.1).

3.4.3.2 Sand and dust proof. The assembly shall be capable of withstanding a sand and dust velocity of 762 ± 152 meter per minute (mpm) for 6 hours at 25 degrees Celcius (°C) and 762 ± 152 (mpm) for 6 hours at 71°C with no evidence of failure, electrical or mechanical. The electrical output shall conform to 3.4.1 (see 4.6.6.2).

3.4.3.3 Fungus resistance. The assembly shall be capable of withstanding a fungus environment at 93 to 100 percent relative humidity without electrical or mechanical failure. The electrical output shall conform to 3.4.1 (see 4.6.6.3).

3.4.3.4 Shock resistance. The assembly shall be capable of withstanding a saw tooth wave shock pulse of 75 gravity unit (g) peak for a duration of 6 milliseconds (msec), with no evidence of failure, electrical or mechanical. The electrical output shall conform to 3.4.1 (see 4.6.7).

3.4.3.5 Corrosion resistance. The assembly shall be capable of withstanding a 5 percent salt solution spray at a maintained temperature of 35° + 1° - 3°C for a minimum (min) of 48 hours with no evidence of failure, electrical or mechanical. The electrical output shall conform to 3.4.1 (see 4.6.8).

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3.4.3.6 Vibration resistance. The assembly shall be capable of withstanding the vibration profile per figure 1 for a period of two hours in each of three mutually perpendicular axis with no evidence of failure, electrical or mechanical. The electrical output shall conform to 3.4.1 (see 4.6.9).

3.4.3.7 Temperature stability. The assembly shall meet the requirements specified herein at ambient temperatures ranging from -32°C to 52°C and shall show no evidence of failure, electrical or mechanical, after storage at an ambient air temperature ranging from -54°C to -8°C (see 4.6.10 and 4.6.11).

3.5 Finish. Non-functional parts exposed to the environment shall be painted white conforming to TT-E-489 or MIL-L-52043 (see 4.6.2).

3.6 Identification marking. The blasting machine shall be marked in accordance with MIL-STD-130 and shall include at least the following (see 4.6.2):

a. Name of item.
b. Manufacturer's name or trademark.
c. Manufacture's part number.
d. Military part identification number (PIN).
e. Manufacturer's serial number.

3.7 Workmanship. Workmanship shall conform to MIL-STD-454, requirement 9 (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective materials.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
   1. Preproduction inspection (see 4.3.1).
   2. Initial production (see 4.3.2).
b. Quality conformance inspection (QCI) (see 4.4).
   1. Examinations (see 4.4.2).
   2. Tests (see 4.4.4).
c. Control tests (see 4.5).

4.3 First article inspection. First article inspection shall be performed on preproduction and initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply assemblies that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first articles sample shall be subject to the approval of the contracting officer.

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two units. Preproduction inspection shall consist of inspection as specified in table II.

4.3.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two units from the first ten produced under the production contract for initial production inspection. the initial production sample shall be inspected as specified in table II.

4.3.3 First article test sequence. First article tests shall be conducted on the sample in accordance with the test sequence specified in table III.

4.3.4 First article inspection failure. Deficiencies found during, or as a result of, the first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article inspection shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

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TABLE II. Classification of inspection.

Title	Requirement	Inspection	First article		QCI		Control 1/ Tests
			Preproduction	Initial production	Exam	Test (100%)	
Group A:							
Materials, design and construction	3.2, 3.2.1, 3.2.2, 3.2.3, 3.3	4.6.1	X	X			
Defects	3.5, 3.6, 3.7	4.6.2	X	X	X		
Group B (Performance):							
Electrical output	3.4.1	4.6.4.1	X	X			X
Electrical output (100%) inspection		4.6.4.1.1				X	
Operating position	3.4.1.1	4.6.4.2	X	X			X
Dielectric strength	3.4.1.2	4.6.4.3	X	X			
Insulation resistance	3.4.1.3	4.6.4.4	X	X			
Rotation	3.4.2.1	4.6.5.1	X	X			
Torque and operational stability	3.4.2.2 and 3.4.2.3	4.6.5.2	X	X			
Group C (Environmental):							
Waterproofness	3.4.3.1	4.6.6.1	X	X			X
Sand and dust proof	3.4.3.2	4.6.6.2	X	X			
Fungus resistance	3.4.3.3	4.6.6.3	X				
Shock resistance	3.4.3.4	4.6.7 and 4.6.7.1	X	X			X
Corrosion resistance	3.4.3.5	4.6.8	X	X			
Vibration	3.4.3.6	4.6.9 and 4.6.9.1	X	X			X
High temperature	3.4.3.7	4.6.10	X	X			X
Low temperature	3.4.3.7	4.6.11	X	X			X

1/ After four successive control tests without a failure, control test shall change in frequency from 1/50 to 1/200.

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TABLE III. First article test sequence. 1/

Title	Requirement	Test	Preproduction sample		Initial production sample	
			1	2	1	2
Design and construction	3.3	4.6.1	X	X	X	X
Performance	3.4	4.6.4	X	X	X	X
Electrical output	3.4.1	4.6.4.1	X	X	X	X
Physical characteristics	3.4.2	4.6.5	X	X	X	X
Waterproofness	3.4.3.1	4.6.6.1	X	X	X	X
Sand and dust proof	3.4.3.2	4.6.6.2	X		X	
Fungus resistance	3.4.3.3	4.6.6.3		X		
Shock resistance	3.4.3.4	4.6.7	X	X	X	X
Corrosion resistance	3.4.3.5	4.6.8	X		X	
Vibration resistance	3.4.3.6	4.6.9	X	X	X	X
High temperature	3.4.3.7	4.6.10	X	X	X	X
Low temperature	3.4.3.7	4.6.11	X	X	X	X

1/ Perform tests in the order listed.

4.4 QCI. QCI shall include the examination of 4.4.2 and the test of 4.4.4. Noncompliance with any of the specified requirements in section 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.4.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

4.4.1.1 Lot formation. An inspection lot shall consist of all assemblies of a single type, submitted at the same time for acceptance.

4.4.1.2 Sample. The sample for (QCI) examination and test shall be randomly selected form the inspection lot in accordance with table IV.

4.4.2 Examination. The sample selected in accordance with 4.4.1.2 shall be examined and defects classified as specified in table V (see 4.6.2). The acceptance number in all cases is zero.

4.4.3 Failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired on corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate form new lots and shall be clearly identified as resubmitted lots.

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TABLE IV. Sampling plan for QCI.

QCI sampling plan		
Inspection lot size	Sample size	
	Examination	
	Major	Minor
2 to 8	5	2
9 to 15	5	2
16 to 25	5	2
26 to 50	5	5
51 to 90	7	5
91 to 150	11	6
151 to 280	13	7
281 to 500	16	9
501 to 1200	19	11
1201 to 3200	23	13
3,201 to 10,000	29	15
10,001 to 35,000	35	15
35,001 to 150,000	40	15
150,001 to 500,000	40	15
500,001 and over	40	15

TABLE V. Classification of defects.

Category	Defects	Requirement	Method of inspection
Critical	None		
Major			
101	Identification markings not as specified.	3.6	Visual
102	Faulty workmanship	3.7	Visual
Minor			
201	Improper finish.	3.5	Visual

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4.4.4 Test (100% inspection). Each item shall be subjected to the (QCI) test specified in table II.

4.4.4.1 Test failure. Failure of the item to pass the QCI examination and tests shall be cause for the Government to refuse acceptance of the item. Any item containing one or more defects either shall not be submitted for Government acceptance or shall be corrected and re-examined or retested and resubmitted without defects for Government acceptance. Resubmitted items shall be kept separate from new items, and shall be clearly identified as resubmitted items.

4.5 Control tests. Unless otherwise specified (see 6.2), control test shall be conducted by the supplier for maintaining control of those requirements whose length, expense, or destructiveness prohibit their inclusion under lot-by-lot inspection. Control test assemblies shall be selected at frequencies specified in 4.5.1 and 4.5.2 and subjected to inspections to verify conformance to the requirements listed in table II. Control test is not required on the initial production control quantity.

4.5.1 Control test of one in fifty (1/50). Except as noted in 4.5, control tests as specified in table II shall be performed on one assembly from each fifty produced or on one assembly every two months, which ever occurs first. The assembly shall be randomly selected from the production quantity represented.

4.5.1.1 Failure of one in fifty. Failure of the assembly to pass any of the specified control test shall be cause for the Government to refuse acceptance of the production quantity represented, when cause for failure is determined, necessary corrections shall be made and proven by test of three randomly selected assemblies from that production quantity. One of the three assemblies shall be subjected to all specified tests. Control test verification for the remaining two assemblies shall be specified to those parameters directly related to the failure cause and those parameters affected by the corrective action taken.

4.5.2 Control test frequencies other than one in fifty. Control tests shall be performed on one assembly at the intervals listed in table II. The assembly may be selected from either the first ten units produced for the interval defined, or the last ten assemblies produced from the preceding interval.

4.5.2.1 Failure of other than one in fifty. Failure of a control assembly to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Method of inspection.

4.6.1 Materials, design and construction. Conformance to 3.2, 3.2.1, 3.2.2, 3.2.3, and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and

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materials conform to requirements. Applicable records shall include drawing, specification, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.5, 3.6 and 3.7 shall be determined by examination for defects listed in table V. Examination shall be visual.

4.6.3 Test fixture. When specified herein, the blasting machine assembly shall be mounted in a test fixture as illustrated in figure 2.

4.6.4 Performance tests.

4.6.4.1 Electrical output. To determine conformance to 3.4.1, the assembly shall be mounted in the test fixture and connected to the resistive loads of table I and subject the rotor handle to the corresponding specified torques. Measure the peak output voltage to verify that it is not less than the specified value (see figure 2).

4.6.4.1.1 Electrical output (100% inspection) (not applicable to PIN 12282970). To determine conformance to 3.4.1 (condition D), each assembly shall be mounted in the test fixture and connected with a 1.47Ω ± 1% resistive load across the output terminals as described in table I, condition D. The electric output shall be minimum 3.8 volts at peak and 1.1 volts, 50 milliseconds later with a rotor shaft input torque not greater than 11.3 newton-meter applied throughout the 150 ± 5° shaft rotation.

4.6.4.2 Operating position. To determine conformance to 3.4.1.1, the assembly shall be mounted in the test fixture and subjected to the electrical output test of 4.6.4.1 at each specified operating position.

4.6.4.3 Dielectric strength. To determine conformance to 3.4.1.2, perform the dielectric strength test between the case and output leads in accordance with MIL-STD-202, Method 301, using 500 volts rms at a frequency of 60 Hz for a period of 60 ± 5 seconds. During this test, verify that the assembly shows no evidence of loosening, cracking, charring, or arcing.

4.6.4.4 Insulation resistance. To determine conformance to 3.4.1.3, measure the insulation resistance between each conductor and case in accordance with MIL-STD-202, Method 302, Test Condition B. Verify that the resistance is not less than 100 megohms.

4.6.5 Physical characteristics.

4.6.5.1 Rotation. To determine conformance to 3.4.2.1, the assembly shall be mounted in the test fixture and the direction and angular rotation of the handle shall be verified.

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4.6.5.2 Torque and operating stability. To determine conformance to 3.4.2.2 and 3.4.2.3, the assembly shall be mounted in the test fixture and subjected it to 10,000 cycles of operation at the rate not in excess of 10 cycles per minute. Perform the test as as follows:

a. 990 cycles at condition B.
b. 10 cycles at condition H.
c. Perform the electrical output test of 4.6.4.1.
d. Repeat for a total of 10,000 cycles.

4.6.6 Environmental tests.

4.6.6.1 Waterproofness. To determine conformance to 3.4.3.1, submerge the assembly in distilled water contained in a vacuum vessel. Observe for air leakage while holding a vacuum differential of 41.4 kPa for not more than 30 minutes. Repeat the test after inverting the assembly. A steady stream of bubbles from any surface shall be cause for rejection. At the conclusion of this test, verify that the assembly passes the electrical output test of 4.6.4.1.

4.6.6.2 Sand and dust proof. To determine conformance to 3.4.3.2, subject the assembly to the sand and dust proof test of MIL-STD-202, Method 110, test condition A, using a sand and dust velocity of 762 ± 152 mpm for 6 hours at 25°C and 762 ± 152 mpm for 6 hours at 71°C. At the conclusion of this test, verify that the assembly passes the electrical output test of 4.6.4.1.

4.6.6.3 Fungus resistance. To determine conformance to 3.4.3.3, subject the assembly to the fungus resistance test of MIL-F-13927, Method A, Class 2. At the conclusion of this test, verify that the assembly passes the electrical output test of 4.6.4.1.

4.6.7 Shock resistance. To determine conformance to 3.4.3.4, subject the assembly to the shock test specified in MIL-STD-810, Method 516, procedure I, for a total of 18 sawtooth wave shock pulses. Peak amplitude shall be 75.0 ± 7.5g with a time duration of 6.0 ± 0.6 msec measured at the 10 percent amplitude points. At the conclusion of this test, verify that the assembly passes the electrical output test of 4.6.4.1.

4.6.7.1 Control test. This test shall be the same as 4.6.7 except the shock level shall be limited to 30 ± 3g.

4.6.8 Corrosion resistance. To determine conformance to 3.4.3.5, subject the assembly to the corrosion resistance test of MIL-STD-202, Method 101, test condition B. At the conclusion of this test, verify that the assembly passes the electrical output test of 4.6.4.1.

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4.6.9 Vibration resistance. To determine conformance to 3.4.3.6, subject the assembly to the vibration test specified in MIL-STD-810, Method 514, Procedure VIII, except vibration level shall be shown in figure I. Sweep time shall be 16 minutes per cycle and the total cycling time per axis shall be two hours in each of the three mutually perpendicular axes. Resonance search shall be conducted at room temperature. After resonance (if any) has been determined place the assembly in a 52°C ambient environment and subject it to vibration sweeps with resonance, dwell at points (if any) detected. At the conclusion of this test, return the assembly to room temperature and verify that it passes the electrical output test of 4.6.4.1.

4.6.9.1 Control tests. This test shall be the same as 4.6.9 except the dwell time at each resonance shall be two minutes and the test shall be conducted at 21 ± 6°C ambient temperature.

4.6.10 High temperature. To determine conformance to 3.4.3.7, subject the assembly to the high temperature test specified in MIL-STD-810, Method 501, Procedure II, except storage temperature of step 4 shall be 71°C. At the conclusion of the storage temperature test, stabilize the assembly at 52°C and subject it to the electrical output test of 4.6.4.1. After testing, return the assembly to room ambient temperature and verify that it passes the electrical output test of 4.6.4.1.

4.6.11 Low temperature. To determine conformance to 3.4.3.7, subject the assembly to the low temperature test specified in MIL-STD-810, Method 502, Procedure I. Maintain the temperature at -54°C for a period of 12 hours minimum. At the conclusion of this time, stabilize the assembly at -32°C and subject it to the electrical output test 4.6.4.1. After testing, return the assembly to room ambient temperature and verify that it passes the electrical output test of 4.6.4.1.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The blasting machine covered by this specification is for emergency use as an alternate firing method in the event of power failure in the vehicle. It is a manually operated device which produces an electrical output through generator action.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable drawing (see 3.3).
c. Whether first article is required (see 3.1).
d. Arrangements for first article inspection and rights of the Government (see 4.4.1 and 4.4.2, and 6.3).
e. If sampling plan for QCI is other than as specified (see 4.4.1).
f. If the control test shall be other than as specified (see 4.5).
g. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

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e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

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o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.3).

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows, or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Subject term (key word) listing.

Corrosion resistance
Electrical output
Fungus resistance
Operational stability
Recovered material
Rotation
Sand and dust proof
Shock resistance
Torque
Vibration
Waterproof

6.6 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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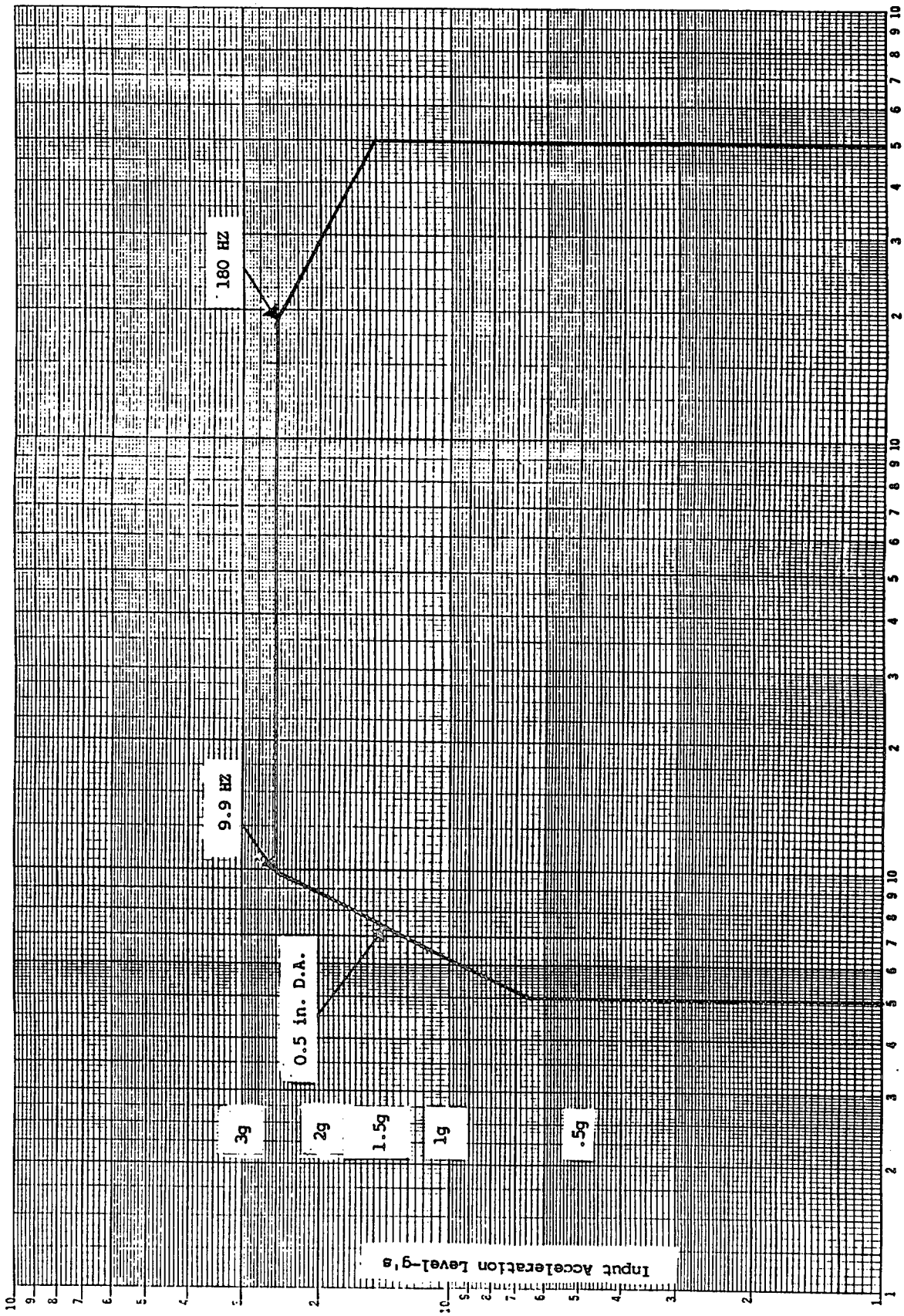


Figure 1. Vibration profile

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COMPONENT METHODS

1. Drive Motor
2. Electric Clutch
3. Cam
4. Switch
5. Constant Torque Springs
6. Timer
7. Holding Fixture
8. TOrque Sensor
9. X-Y Plotter
10. Driver

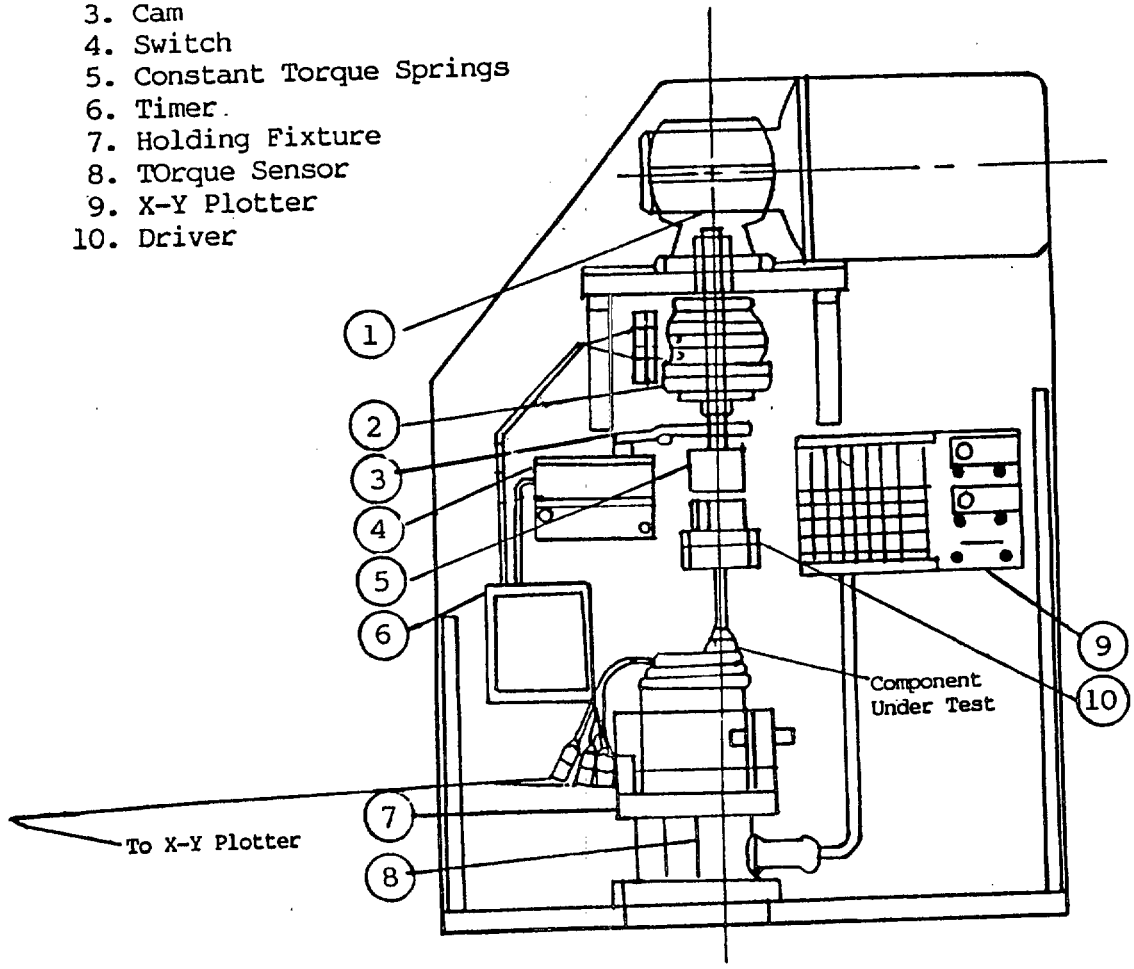


Figure 2. Test fixture diagram.

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FSC 2590

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